Design Briefs

December 2025

1 Prototype

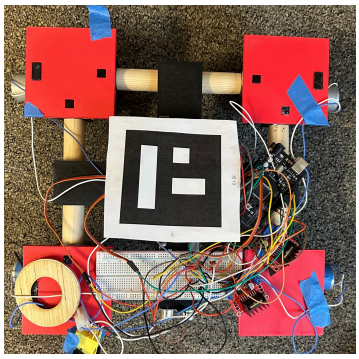


Figure 1: OTV sky view

` Begin by 3D printing the four platform pieces that form the four corners of the OTV as seen in figure 1. Each platform should contain three holes: two square holes and one rectangular hole. Each hole must be centered on a side length of the platform. The two square holes must be on adjacent sides (not opposite one another), and the rectangular hole should oppose one of the square holes that will act as a receptor for other 3D-printed components. Next, 3D print eight cube-shaped connector pieces with a square extrusion designed to snap tightly into the square holes on the corner platforms. Each cube should be hollow on one side to allow a cylindrical piece to be inserted. Hot glue the face of each hollowed cube that contacts the platform, then snap it into place. Repeat this process for all eight cubes. Retrieve four wooden cylinders of equal length and diameter, sanding the ends until they fit tightly into the hollowed cubes. Insert the cylinders into the cubes to connect all four platforms and form the main frame of the OTV.

Next, 3D print a plus-shaped cross with a circular center and a large cube-shaped opening to accommodate the full pinion-rack system. The cross should be sized so that each arm rests directly over a wooden cylinder. To mount the cross, 3D print four rectangular prism pieces with one long side open so they can sit on top of the cylinders, with a flat top surface. Hot glue the flat tops of these pieces to the ends on the underside of the plus-shaped cross, then hot glue each cylinder at the contact point between the cylinder and the cross to secure the structure.

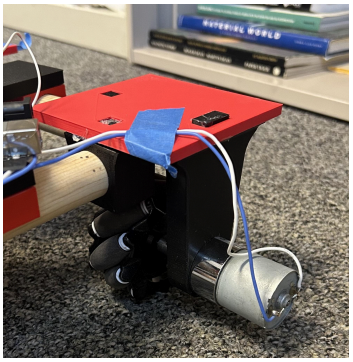


Figure 2: Wheel and motor alignment

For the wheel assemblies seen in figure 2, 3D print four elongated components with a rectangular extrusion on the flat side that fits into the rectangular hole on each platform, and a circular cavity on the bottom to hold a motor. Hot glue these pieces to the platforms on the flat side contacting the corners, ensuring the rectangular extrusion locks into the rectangular hole. Orient each piece so the motor faces outward and the wheel sits inside or directly beneath the OTV frame. Attach omni wheels to the motors by pushing them onto the shafts and securing them with screws.

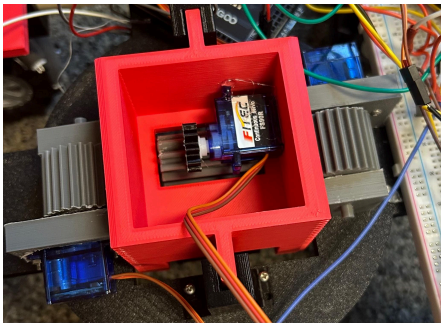


Figure 3: Pinion racquet system

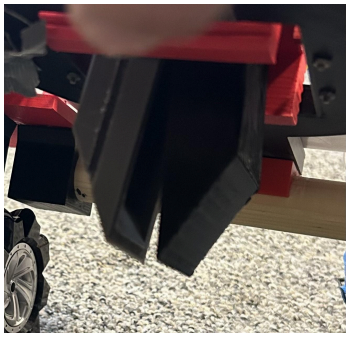


Figure 4: the claw

For the pinion-rack claw system in figure 3, 3D print two pillar-like structures with expanded bases and a longitudinal slit that acts as a guide rail for the claw's box base. Then 3D print the claw box base, the claw itself in two pieces in figure 4, the gears, and a servo holder. On the sides containing the guide rails, hot-glue the gear system and servo holders in place, insert the servos, and slide the box base into the guide rails so it meshes with the gear system. Hot glue the stationary portion of the claw to the base of the box, and attach the moving portion to the gear connected to another servo mounted inside the box.

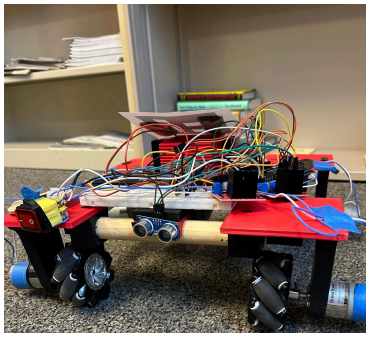


Figure 5: Front view

Finally, attach proximity sensors in desired locations (one was placed at the front as seen in figure 4), and tape a camera to the bottom of the OTV at a slight inward angle, aimed toward the claw's pickup point.

2 Mission Details

Our team-specific mission included three objectives: determine which out of four cups in a two-by-two plot contains a plantable substrate, plant a seed at least 1 cm deep into the plantable substrate, and collect at least a 10g sample of the non-plantable substrate. After collecting the substrate, it had to proceed with the rest of the mission by successfully navigating through obstacles.

In order to accomplish these tasks, our team decided to create a claw mechanism attached to a rack and pinion system located in the middle of our OTV (Figure 6). This entire mechanism was 3D-printed using PLA plastic with the exception of the motors and wires. The plastic body was printed

into several parts: a hollow cube-shaped body with grooves acting as the teeth for the rack, one scooper that remains static, and one moveable scooper with a toothed butt (Figure 7).

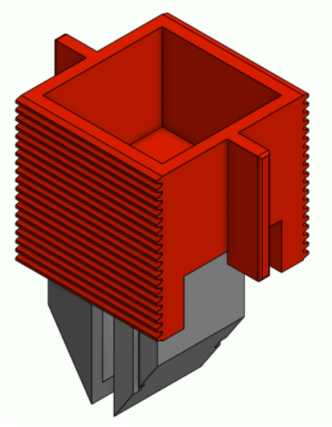


Figure 6: CAD screenshot of claw assembly

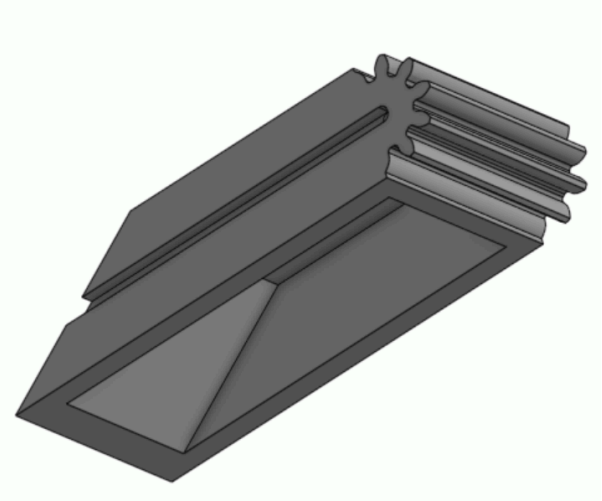


Figure 7: Moveable scooper with gear on backside

The static scooper is hot-glued onto the bottom of the box, while the rotating scooper is held in place by two short pegs that slide into the scooper's slots, allowing it to rotate freely. The entire claw assembly would slide into a square hole in the middle of our chassis with two vertical brackets to improve stability, as seen in figure 8. Both toothed sides of the box make contact with two gears, each powered by one servo motor on opposite sides. This allows the claw to move up or down, as when both gears spin outwards or inwards, the claw will move up or down respectively. One servo motor is attached to the inside of the box, powering a single gear that allows the scooper to rotate.

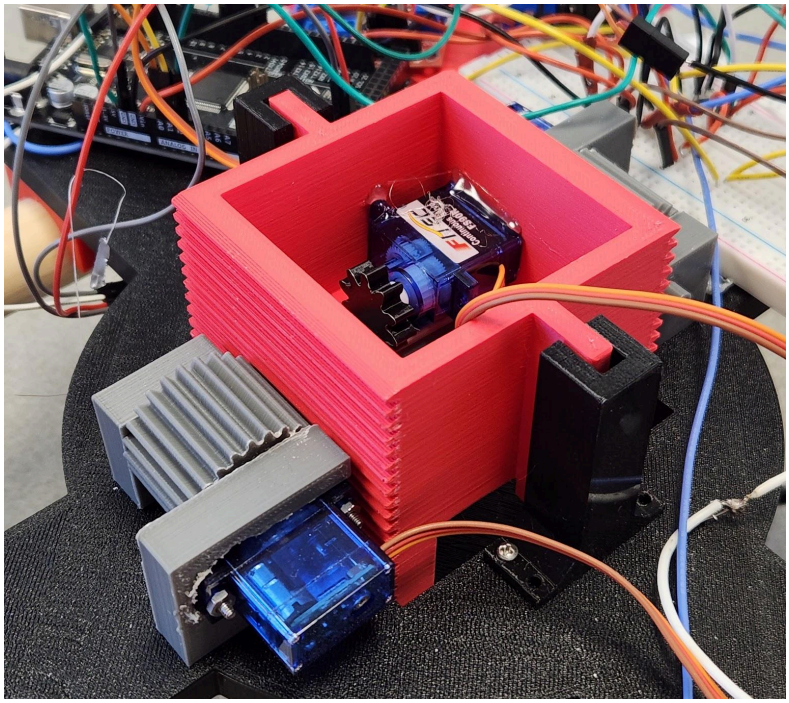


Figure 8: Claw assembly fully integrated into OTV

Our specific objectives also rely on a Wi-Fi camera located on the bottom of our OTV's chassis (Figure 9). This camera is utilized by an AI model we trained to differentiate the plantable and non-plantable substrates. When the camera is oriented on top of one plot, the model will return one of two arbitrary values corresponding to a plantable or non-plantable substrate.

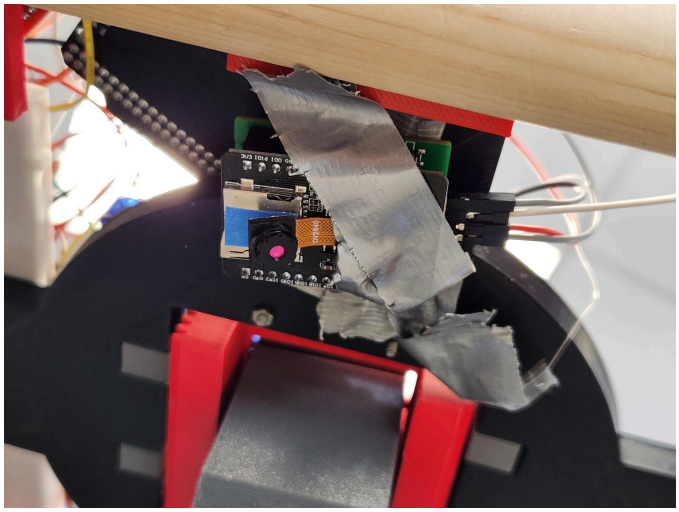


Figure 9: Wi-Fi camera used to find the plantable substrate

With these systems combined, our OTV would complete these objectives by moving over to our mission site until the camera is oriented over one plot and scan to see if the substrate is plantable. If it is plantable, the claw would come down and plant the seed, and would continue to the next plot to pick up the non-plantable substrate. If not, the OTV would move on to the next plot and repeat the same process. A more in-depth flowchart of this algorithm can be found below (Figure 10).

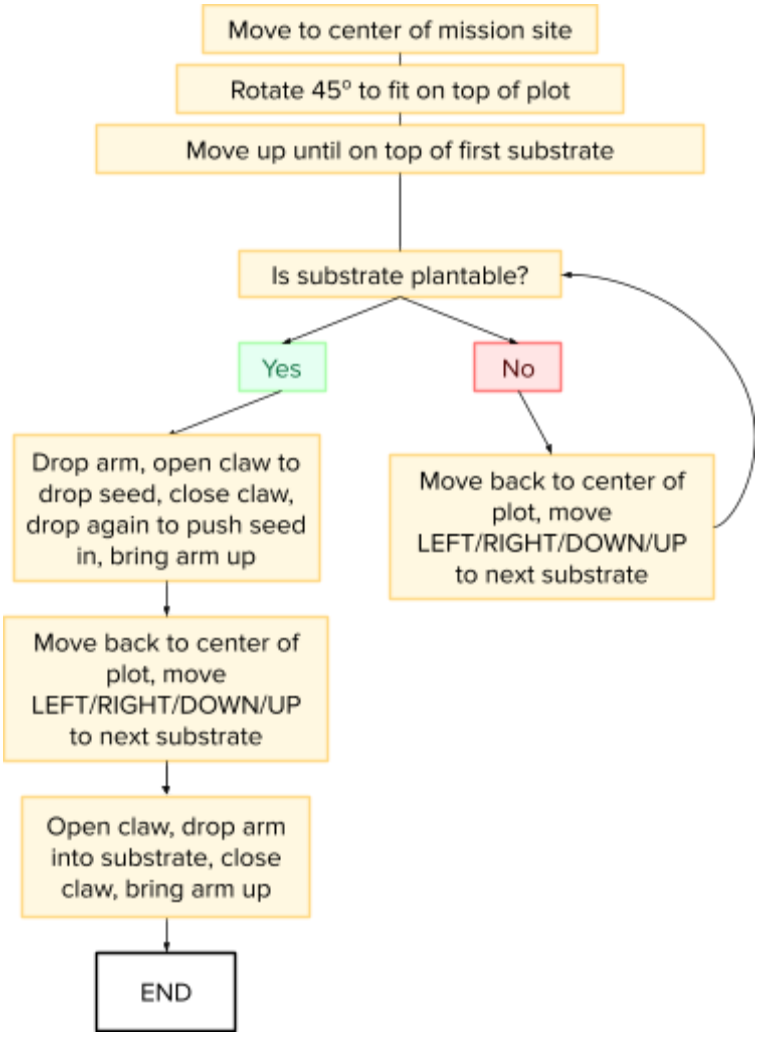


Figure 10: Flowchart of control algorithm

Furthermore, our team also had to complete several other universal objectives, including navigating to our mission site, navigating past three randomly placed obstacles, and navigating into the destination zone of the 4 m by 2 m arena. In order to accomplish these tasks, we utilized the vision system to track where our OTV's ArUco marker was on the arena, as well as a proximity sensor taped to the front of our OTV to sense obstacles in front of it (Figure 11). We only needed one proximity sensor because we needed to see what was in front of us, so that we could proceed through the arena without hitting an obstacle in the front.

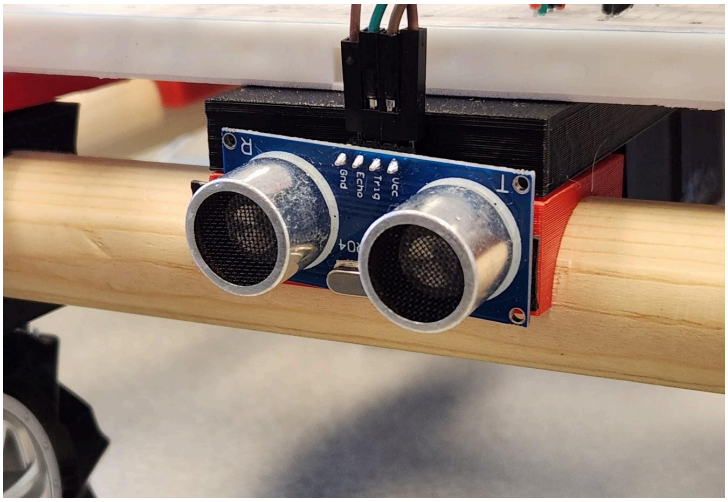


Figure 11: Proximity sensor

If the OTV detected an obstacle, it would move to the right if it was on the top half of the arena ($y > 1$ m) or move left if it was on the bottom half of the arena ($y < 1$ m) until it essentially reached the vertical center of the arena, where it would continue to move forward. Once our OTV passed all of the obstacles, it would move left until it reached about $y = 1.5$ m, which was about the center of the limbo blocking the final destination. Our OTV would then move straight until it passed under the limbo and reached the destination, stopping all function.

3 Propulsion Modeling

For our propulsion system, we needed a configuration that provides both precise maneuverability and sufficient torque to support the OTV's weight and movement over the substrate field. Because our mission involves navigating to specific locations, aligning the arm over substrates with high accuracy, and passing under obstacles, we selected an omni-wheel configuration. Omni-wheels allow for smooth multi-directional movement without requiring complex steering mechanisms, which simplifies both the control system and the physical design of the drivetrain. This choice aligns with our design goals of efficiency and precision.

To determine appropriate motor selection, we first calculated the operating torque. When taking the number of wheels into account, the torque the motor experiences is determined through static equilibrium, assuming the OTV moves at a constant velocity. Using the free-body diagram from figure 12, the following equation can be derived.

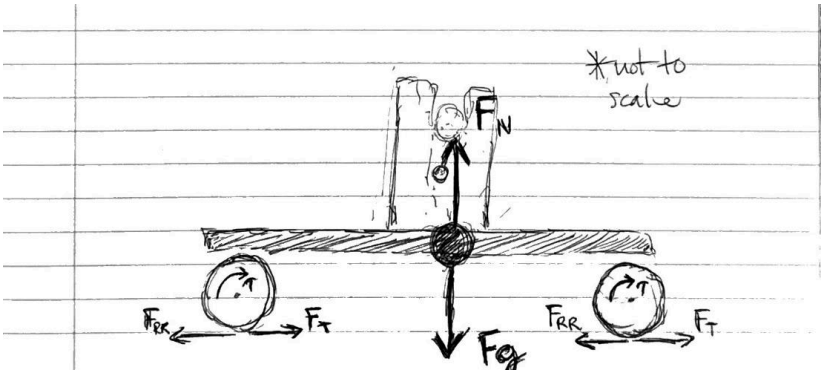


Figure 12: Free body diagram of OTV

$$0 = \sum_{i=0} F_i = 4F_{Friction} - 4F_{Propulsion}$$

$$F_{Friction} = F_{Propulsion}$$

The friction can be derived using the normal force, along with the coefficient of rolling resistance $C_{RR,}$ and the mass m.

$$F_{Friction} = F_N \times C_{RR}$$

$$F_N = \frac{mg}{4}$$

Which can then be used to find the propulsion force and torque on each wheel, using the radius of each wheel, R.

$$\frac{C_{RR}mg}{4} = F_{Propulsion}$$

$$\tau = RFsin(\theta)$$

$$\tau = RF_{Propulsion}$$

$$\tau = \frac{RC_{RR}mg}{4}$$

Using the measured mass of m = 2.56kg, the radius of R = 4.5cm or R=0.045m, and an estimated coefficient of rolling resistance (from data given in ENES100) of C_{RR} = 0.08, the torque can be calculated to be 0.023Nm, which is 0.319oz-in, using the equation $\frac{(0.08)(0.045m)(9.8\frac{m}{s^2})(2.56)}{4}$. This torque is also the OTV's operating torque. To account for worst-case scenarios, the motor chosen has a higher stall torque, significantly higher than the no-load torque and somewhat higher than the stall torque. The motor runs at 12 volts, which can be represented by the torque curve, using the stall torque 83.32oz-in and the no-load speed 30RPM.

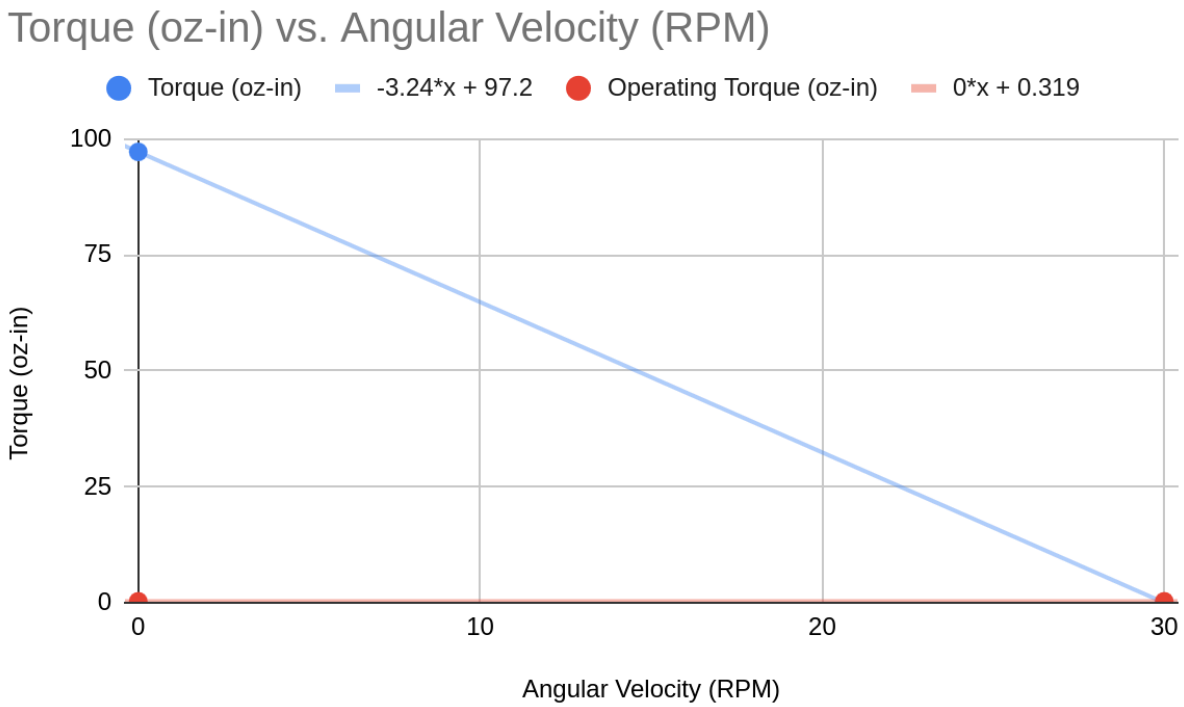


Figure 13: Torque vs angular velocity graph

Using the operating torque, the linear operating speed can be derived. By taking $\omega(\tau_{operating})$, after converting from 30RPM to $\frac{rad}{s}$, the operating angular speed can be found to be $3.141\frac{rad}{s}$. It can then be converted to speed using the formula $v = \omega r$, which, using the radius of r = 0.045m, would be v = $1.45\frac{m}{s}$, the operating linear speed. The current should be driven much lower than the operating current, because $0.141\frac{m}{s}$. Our observed speed was $0.114\frac{m}{s}$, which was extremely close to our estimated speed.

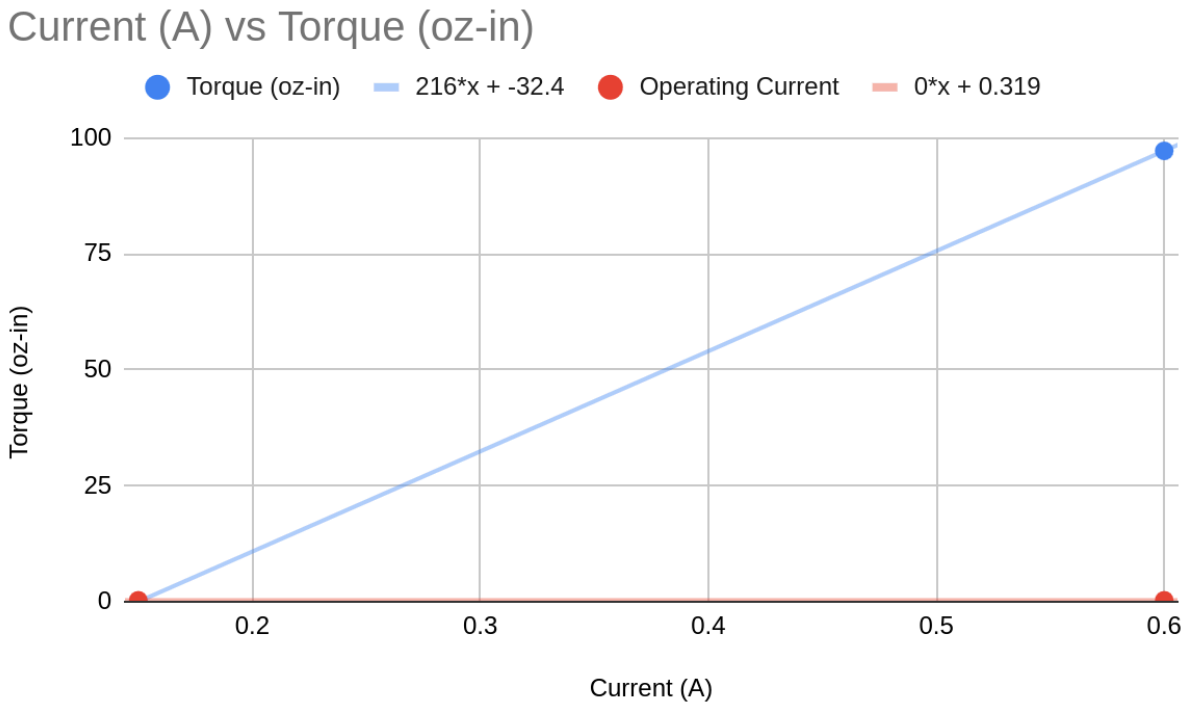


Figure 14: Current vs torque graph

For determining the operating current of each motor, the operation can be found by utilizing a graph of the torque and current, and by finding the current amount for the operating torque. This is the current draw when the motor is running under its operating torque. The graph above, shows the current draw of each motor to be 25mA.

This propulsion configuration was chosen for its light weight, high controllability, and energy efficiency, all of which are critical for meeting both the mission performance requirements and the power budget. Additionally, omni-wheel steering reduces the number of active motors needed, minimizing energy consumption while maintaining precise navigation capabilities.

4 Electronics

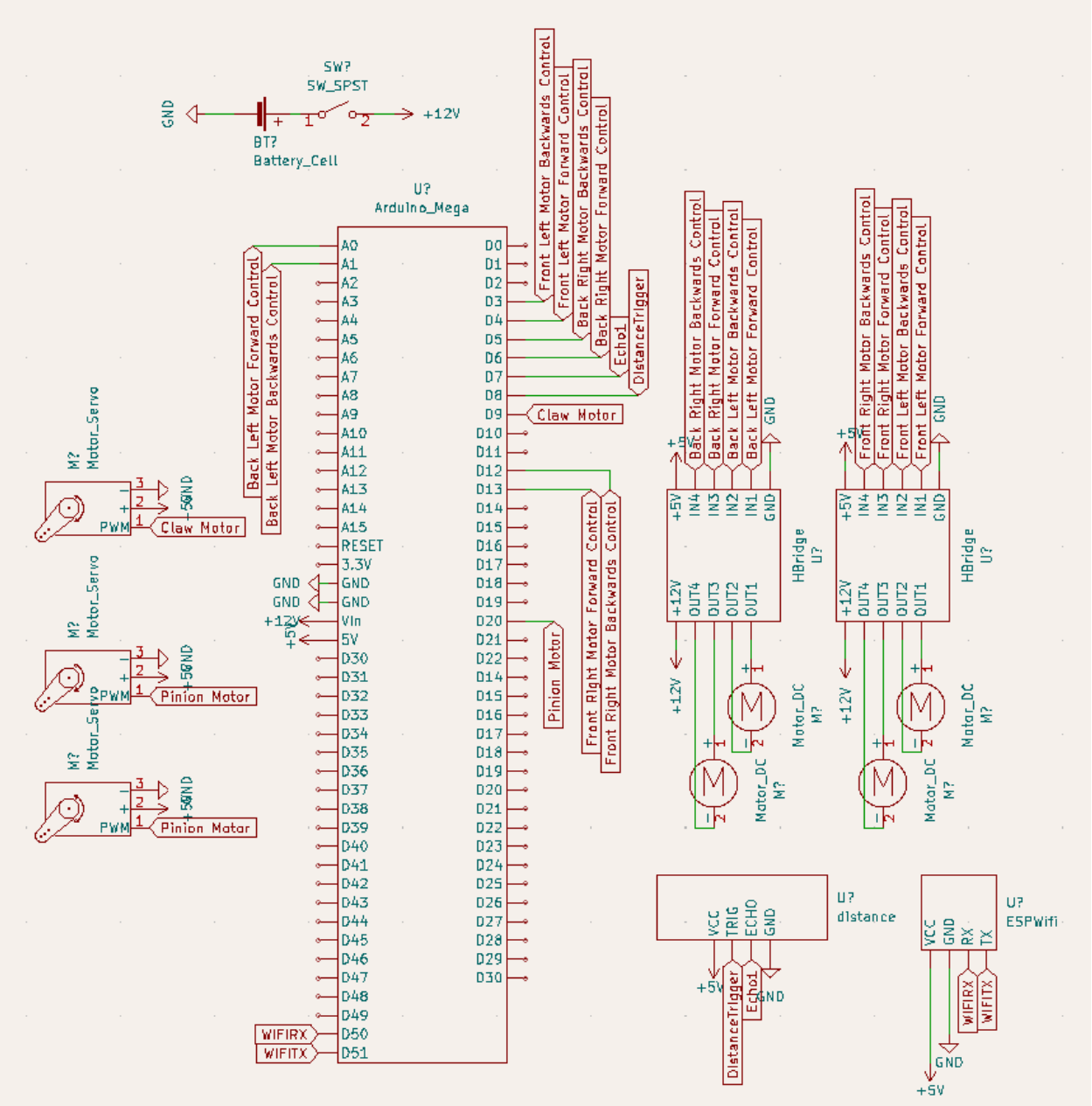


Figure 15: Circuit Schematics

The OTV used a Nickel-Metal Hydride Batter provided by the school store to power the OTV. It was a 12V battery, with 2000mAh, with a maximum recommended discharge rate of 2A. It used 1 ultrasonic sensor, two H-Bridges for navigation, an Arduino Mega as a microcontroller, and three servos for control of the rack and pinion system (Figure 15). The OTV will not have modulatable power for the motors, because different speeds are not necessary for the tasks required. Therefore, PWM pins are not used for driving the H-Bridges, instead using standard digital pins.

The camera that is underneath the OTV right in front of the center hole where our claw will drop the seed. This camera will be used to see the plantable substrate, based on the size of each of the 4 types of substrates' particles. Based on size, it will find the correct substrate. This will be done using machine learning. The OTV wheels will be elevated enough so that it will allow the OTV to go over the substrates, so that the camera can see down on them. If the camera is over a non-plantable substrate, it will send feedback to the robot to use its omni-wheels to either move up, down, left, or right to the next substrate and repeat the process, etc. (e.g. the OTV will move in relation so that the camera also knows the location of the claw).

To calculate the expected energy usage E, first the overall power $\sum P_i$ can be found by summing the power draw of each current using the formula P = V × I where V is the voltage of the chosen component and I is the current draw.

$$\sum_{i=0} (V_i \times I_i) = \sum_{i=0} P_i \tag{5.1}$$

$E = \int Pdt$, which can be simplified E = P × Δt since P is constant. Using this the energy can be calculated with the formula

$$\sum_{i=0} P_i \times \Delta t = E \tag{5.2}$$

Or more explicitly

$$\left[\sum_{i=0} V_i I_i\right] \Delta t = E \tag{5.3}$$

Name	Current (mA)	Voltage (V)	Power ($V \times I$) (W)	Number of Components
Ultrasonic Distance Sensor	2.2	3	0.0066	1
Arduino Uno Rev3	20	12	0.24	1
Motor	25	12	0.24	4
H-Bridge	36	12	0.432	2
Servo	100	5	0.5	3
Wifi Module	negligible	5V	0	1

Table A: Power Chart

Using formula 5.1, the total power can be found to be:

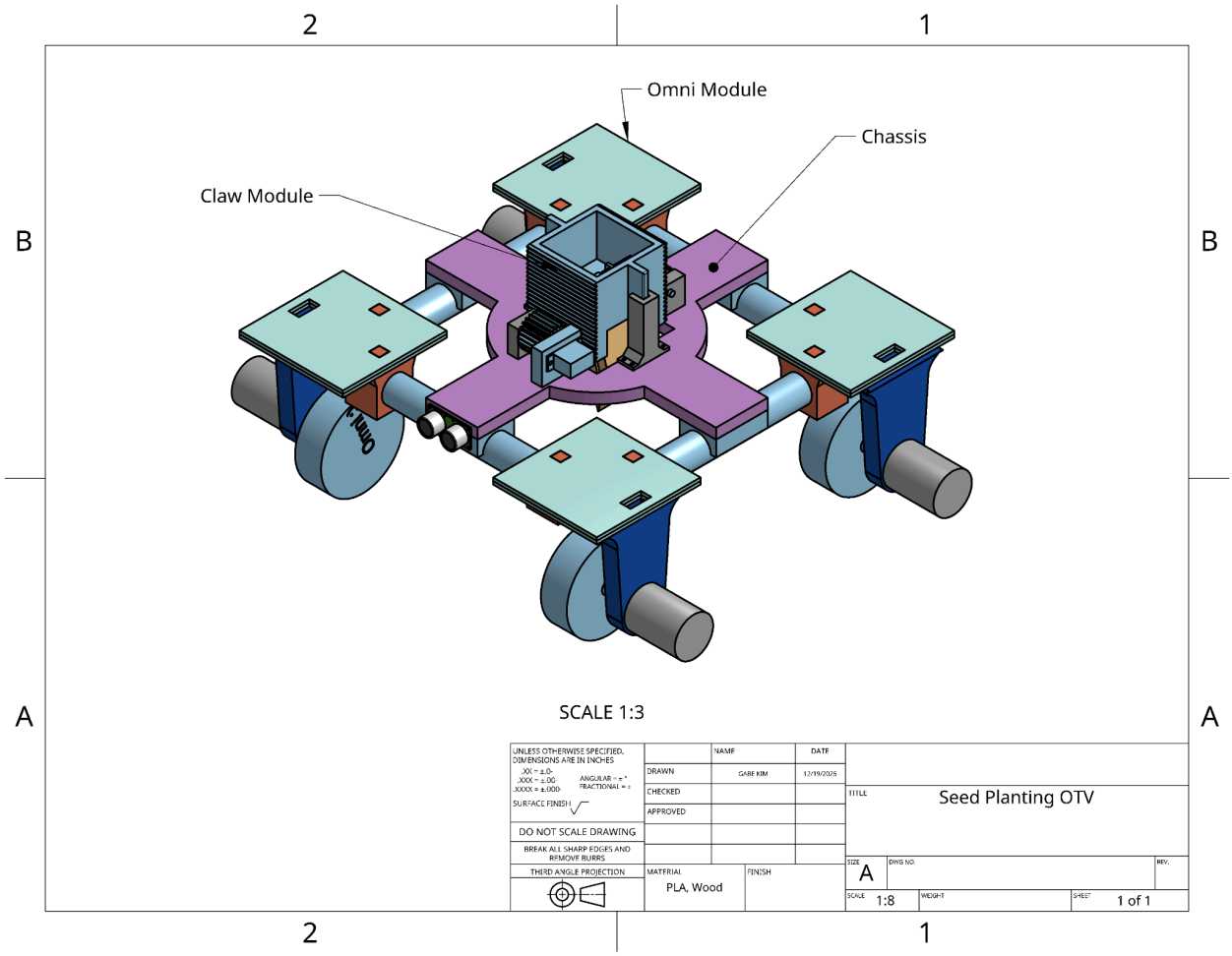
*0.0066*2+0.24+0.24*4+0.432*4+0.5*3+0+0.066=4.07W*

The OTV should run for 20 minutes, which is equivalent to 0.33 hours, which means $\Delta t = 0.33h$. Multiplying Δt by the total power, the battery's Energy requirements can be found to be 1.34Ah. A battery of 2Ah would be more than sufficient. Through this, we decided to use the battery from the school store.

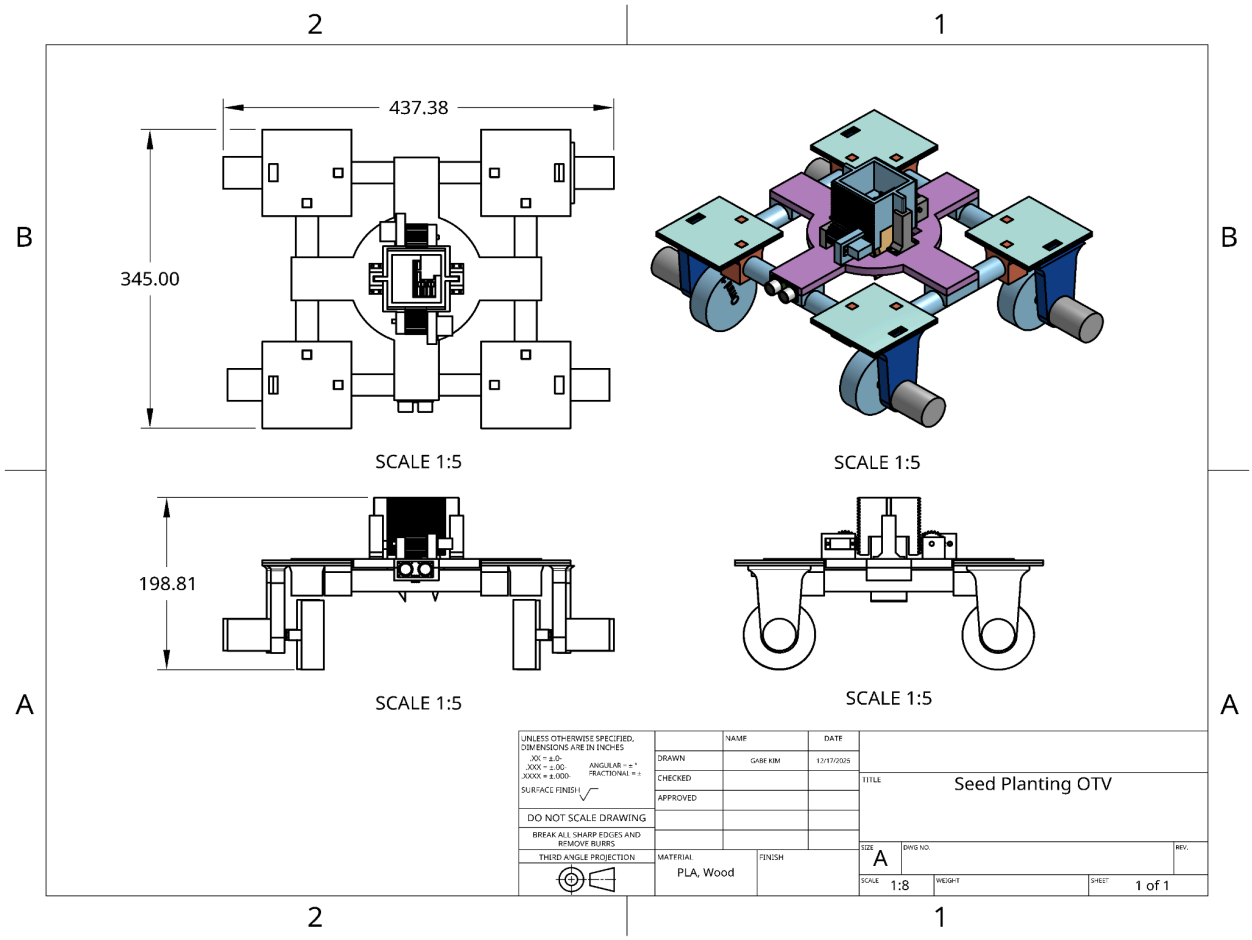
To find the operating current, the current of all components can be summed by all the individual current draws from electrical components:

*2.2+20+25*4+36*2+100+20=**314.2mA***

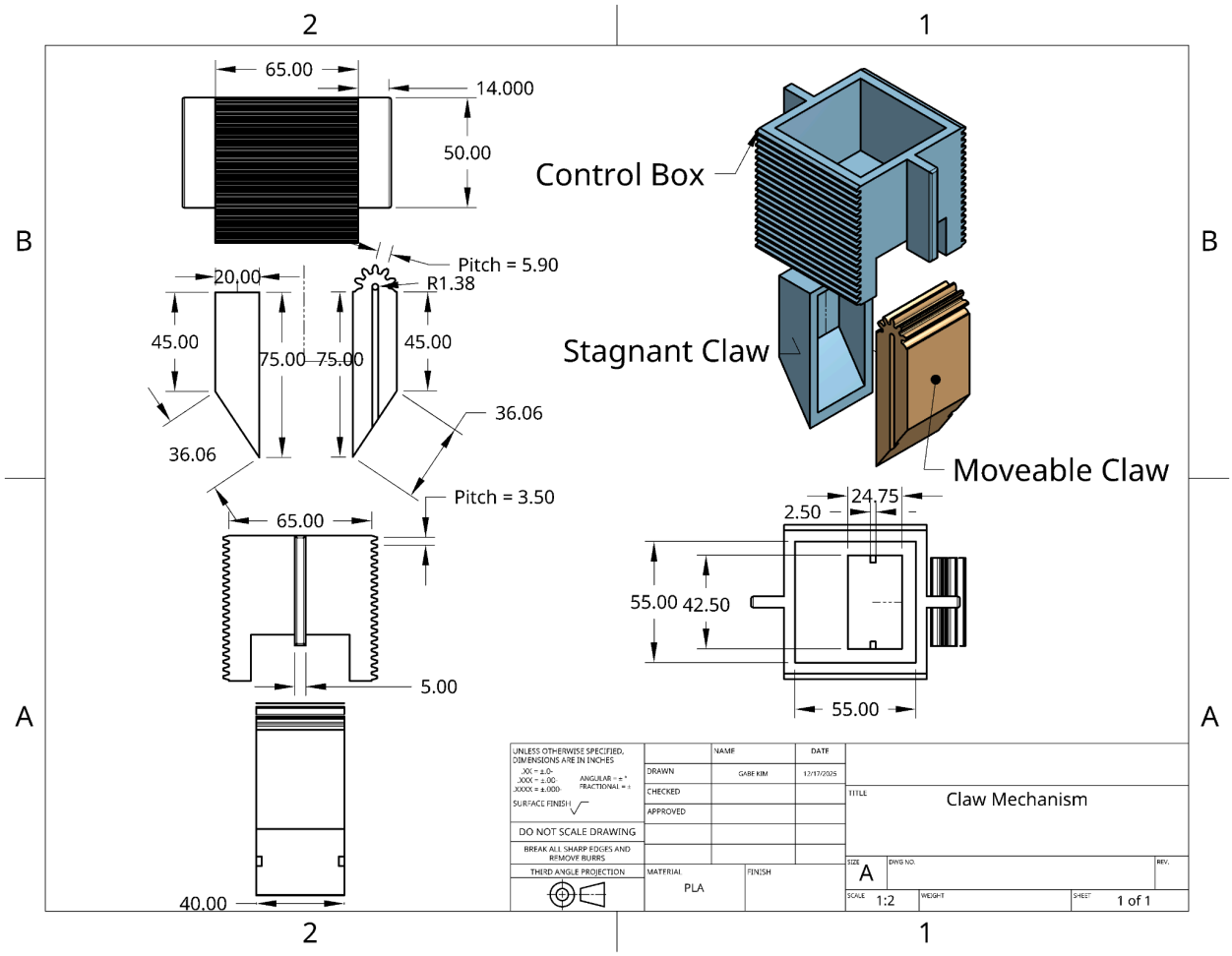
5 Engineering Drawings



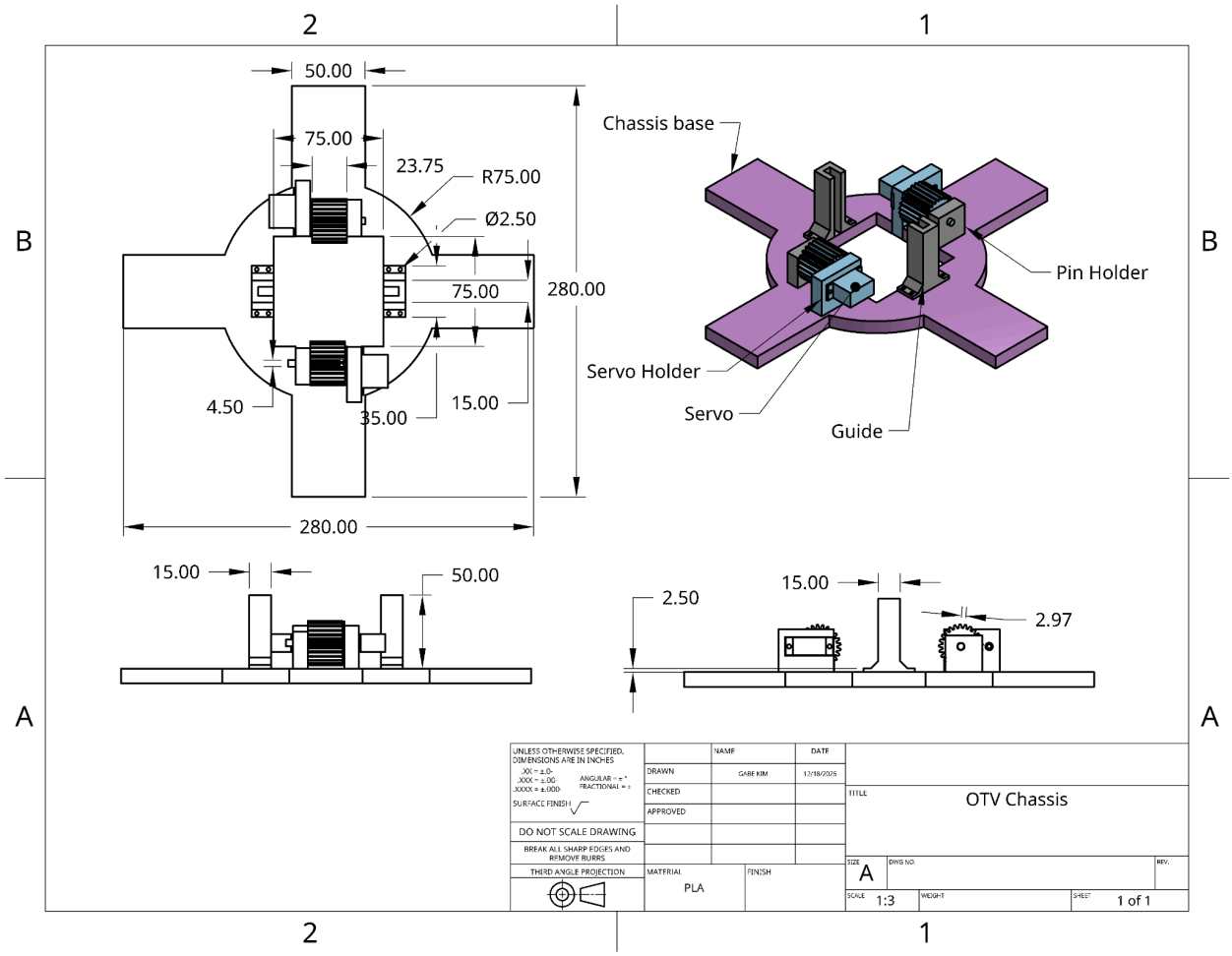
Drawing A: Final OTV Isometric View



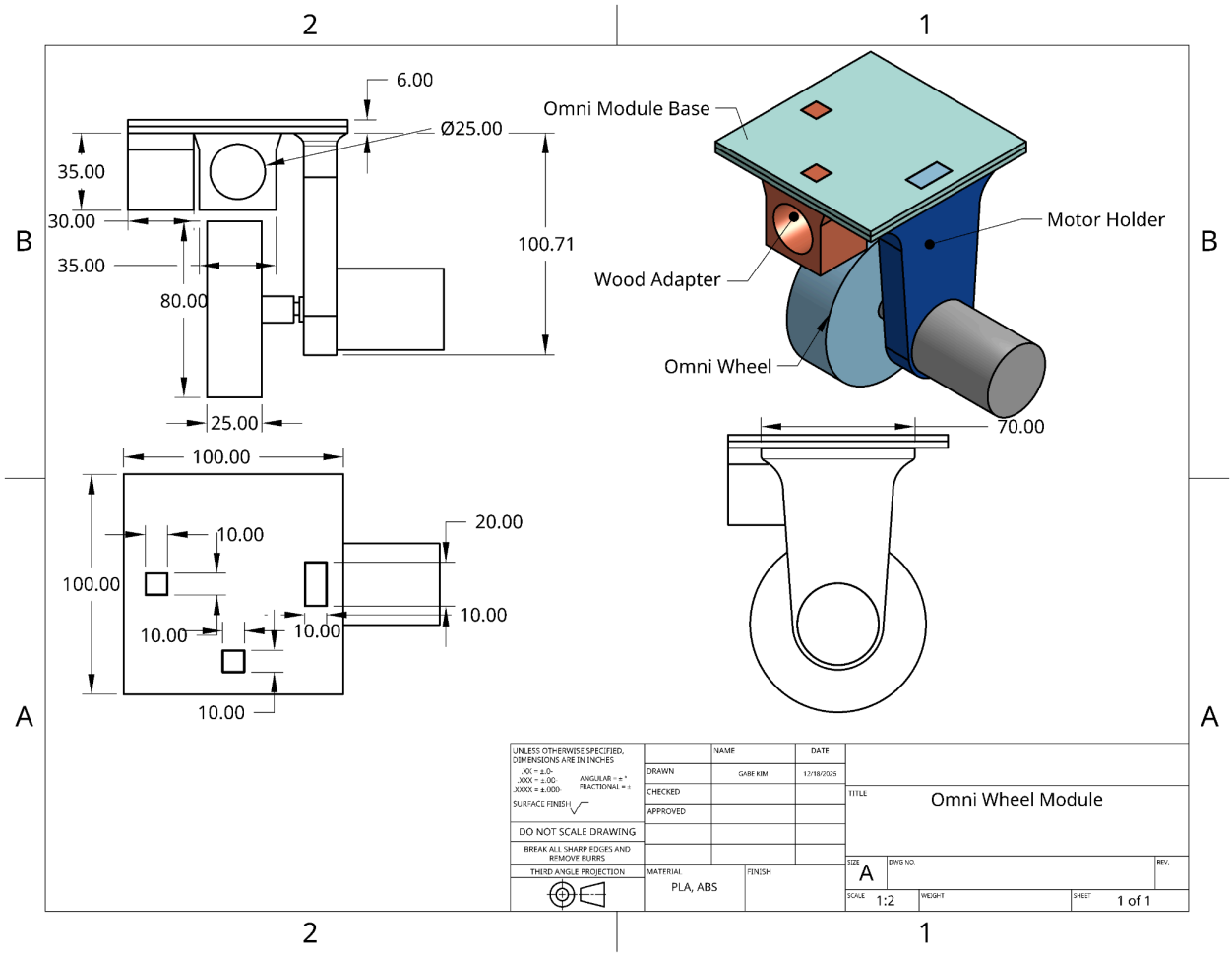
Drawing B: Final Max Dimensioned Assembly of OTV



Drawing C: Claw Mechanism Subassembly



Drawing D: OTV Chassis Subassembly



Drawing E: Omni Wheel Module Subsystem

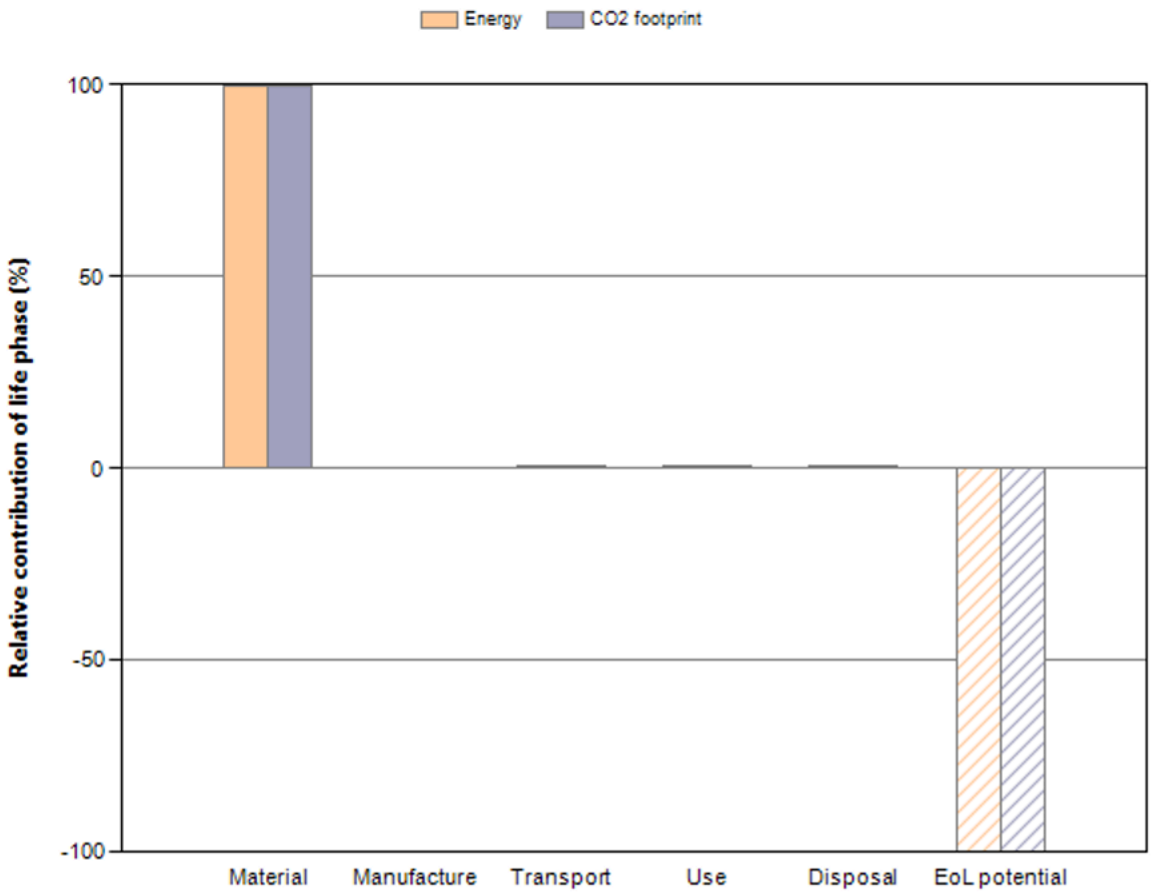
6 Sustainability

Ansys
GRANTA EDUPACK

Eco Audit Report

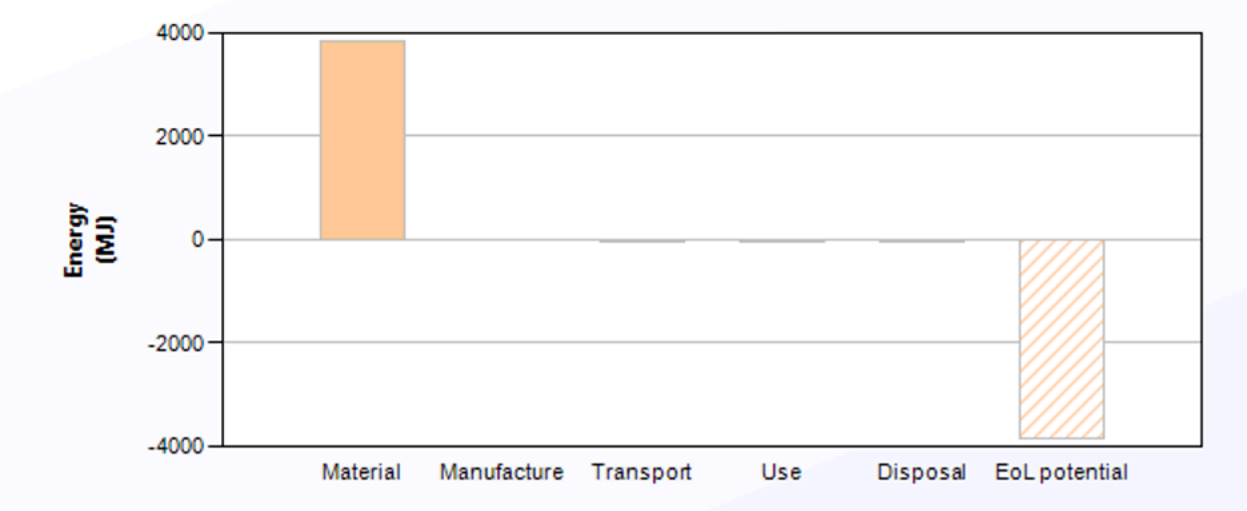
Product name	OTV_BotanicalBuddies
Country of use	North America
Product life (years)	0.25

Summary:



Phase	Energy (MJ)	Energy (%)	CO2 footprint (kg)	CO2 footprint (%)
Material	3.84e+03	99.8	283	99.8
Manufacture	0	0.0	0	0.0
Transport	4.67	0.1	0.336	0.1
Use	1.62	0.0	0.0939	0.0
Disposal	0.57	0.0	0.0399	0.0
Total (for first life)	**3.85e+03**	**100**	**283**	**100**
End of life potential	-3.84e+03		-283	

Energy Analysis:



	Energy (MJ/year)
Equivalent annual environmental burden (averaged over 0.25 year product life):	1.54e+04

Detailed breakdown of individual life phases

Material:

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	Energy (MJ)	%
Kill Switch	Transistor	Reused part	0.05	1	0.05	0	0.0
Battery	NiMH, rechargeable battery (for laptops)	Reused part	0.29	1	0.29	0	0.0
Servos	Fan	Virgin (0%)	0.0023	3	0.0069	1.7	0.0
Motors	Fan	Virgin (0%)	0.2	4	0.78	1.9e+02	5.0
Ultrasonic Sensor	Printed circuit board assembly	Reused part	0.006	1	0.006	0	0.0
Breadboard	Plug, inlet and outlet	Virgin (0%)	0.06	1	0.06	7	0.2
Arduino Mega	Printed circuit board assembly	Virgin (0%)	0.036	1	0.036	4.7	0.1
Wires	Cable	Virgin (0%)	0.005	1	0.005	0.45	0.0
PLA	PLA (general purpose)	Virgin (0%)	0.1	1	0.1	4.6	0.1
Wooden Dowels	Beech (fagus grandifolia) (t)	Reused part	0.19	4	0.75	0	0.0
H-Bridges	Integrated circuit (large)	Virgin (0%)	0.23	2	0.45	3.6e+03	94.5
WifiCam	Printed circuit board assembly	Reused part	0.05	1	0.05	0	0.0
Total				**21**	**2.6**	**3.8e+03**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

Manufacture:

Component	Process	Amount processed	Energy (MJ)	%
Total				**100**

Transport:

Breakdown by transport stage

Stage name	Transport type	Distance (km)	Energy (MJ)	%
Ultrasonic Sensor	40 tonne (6 axle) truck	3.2e+02	0.67	14.4
Servos	Light goods vehicle	46	0.26	5.6
Wheels	55 tonne (8 axle) truck	1e+03	1.9	41.3
H-Bridges	Light goods vehicle	46	0.26	5.6
Motors	32 tonne (4 axle) truck	2.1e+02	0.51	11.0
Breadboard	32 tonne (4 axle) truck	2.1e+02	0.51	11.0
Arduino Mega	32 tonne (4 axle) truck	2.1e+02	0.51	11.0
Total		**2.1e+03**	**4.7**	**100**

Breakdown by components

Component	Mass (kg)	Energy (MJ)	%
Kill Switch	0.05	0.09	1.9
Battery	0.29	0.53	11.3
Servos	0.0069	0.012	0.3
Motors	0.78	1.4	30.1
Ultrasonic Sensor	0.006	0.011	0.2
Breadboard	0.06	0.11	2.3
Arduino Mega	0.036	0.065	1.4
Wires	0.005	0.009	0.2
PLA	0.1	0.19	4.0
Wooden Dowels	0.75	1.3	28.8
H-Bridges	0.45	0.82	17.5
WifiCam	0.05	0.09	1.9
Total	**2.6**	**4.7**	**100**

Use:

Static mode

Energy input and output type	Electric to mechanical (electric motors)
Country of use	North America
Power rating (W)	24
Usage (hours per day)	2
Usage (days per year)	15
Product life (years)	0.25

Relative contribution of static and mobile modes

Mode	**Energy (MJ)**	**%**
Static	1.6	100.0
Mobile	0	
Total	**1.6**	**100**

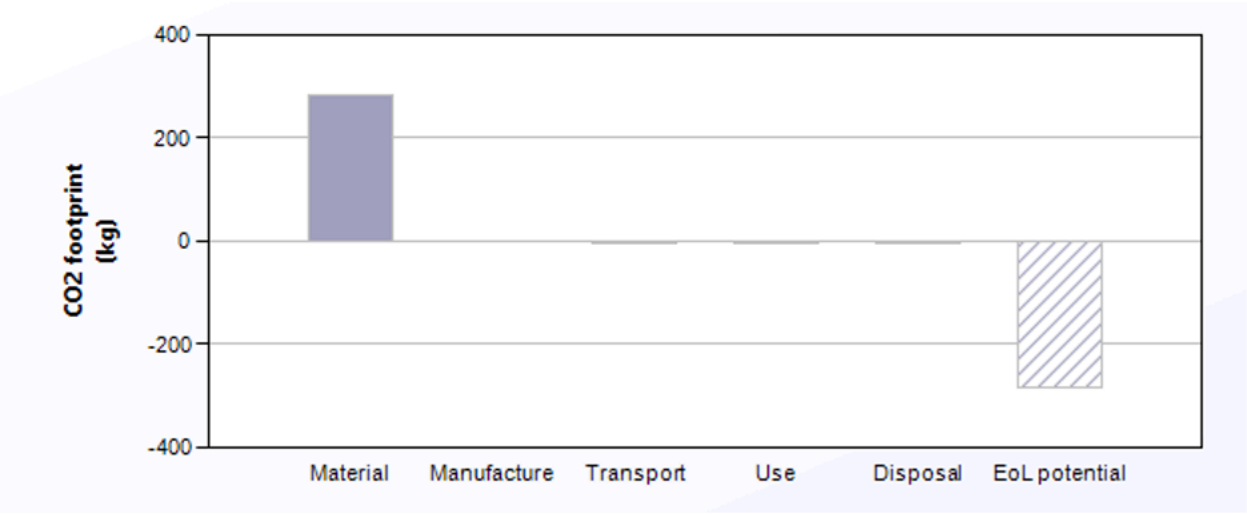
Disposal:

Component	End of life option	Energy (MJ)	%
Kill Switch	Reuse	0.01	1.8
Battery	Reuse	0.059	10.3
Servos	Reuse	0.0014	0.2
Motors	Reuse	0.16	27.4
Ultrasonic Sensor	Reuse	0.0012	0.2
Breadboard	Reuse	0.012	2.1
Arduino Mega	Reuse	0.0072	1.3
Wires	Reuse	0.001	0.2
PLA	Recycle	0.072	12.6
Wooden Dowels	Landfill	0.15	26.2
H-Bridges	Reuse	0.091	15.9
WifiCam	Reuse	0.01	1.8
Total		**0.57**	**100**

EoL potential:

Component	End of life option	Energy (MJ)	%
Kill Switch	Reuse	0	0.0
Battery	Reuse	0	0.0
Servos	Reuse	-1.7	0.0
Motors	Reuse	-1.9e+02	5.0
Ultrasonic Sensor	Reuse	0	0.0
Breadboard	Reuse	-7	0.2
Arduino Mega	Reuse	-4.7	0.1
Wires	Reuse	-0.45	0.0
PLA	Recycle	-3.1	0.1
Wooden Dowels	Landfill	0	0.0
H-Bridges	Reuse	-3.6e+03	94.5
WifiCam	Reuse	0	0.0
Total		**-3.8e+03**	**100**

CO2 Footprint analysis



	CO2 (kg/year)
Equivalent annual environmental burden (averaged over 0.25 year product life):	1.13e+03

Detailed breakdown of individual life phases

Material:

Component	Material	Recycled content* (%)	Part mass (kg)	Qty.	Total mass (kg)	CO2 footprint (kg)	%
Kill Switch	Transistor	Reused part	0.05	1	0.05	0	0.0
Battery	NiMH, rechargeable battery (for laptops)	Reused part	0.29	1	0.29	0	0.0
Servos	Fan	Virgin (0%)	0.0023	3	0.0069	0.081	0.0

Motors	Fan	Virgin (0%)	0.2	4	0.78	9.1	3.2
Ultrasonic Sensor	Printed circuit board assembly	Reused part	0.006	1	0.006	0	0.0
Breadboard	Plug, inlet and outlet	Virgin (0%)	0.06	1	0.06	0.31	0.1
Arduino Mega	Printed circuit board assembly	Virgin (0%)	0.036	1	0.036	0.35	0.1
Wires	Cable	Virgin (0%)	0.005	1	0.005	0.034	0.0
PLA	PLA (general purpose)	Virgin (0%)	0.1	1	0.1	0.23	0.1
Wooden Dowels	Beech (fagus grandifolia) (t)	Reused part	0.19	4	0.75	0	0.0
H-Bridges	Integrated circuit (large)	Virgin (0%)	0.23	2	0.45	2.7e+02	96.4
WifiCam	Printed circuit board assembly	Reused part	0.05	1	0.05	0	0.0
Total				**21**	**2.6**	**2.8e+02**	**100**

*Typical: Includes 'recycle fraction in current supply'

***User-defined material

Manufacture:

Component	**Process**	**Amount processed**	**CO2 footprint (kg)**	**%**
Total				**100**

Transport:

Breakdown by transport stage

Stage name	Transport type	Distance (km)	CO2 footprint (kg)	%
Ultrasonic Sensor	40 tonne (6 axle) truck	3.2e+02	0.048	14.4
Servos	Light goods vehicle	46	0.019	5.6
Wheels	55 tonne (8 axle) truck	1e+03	0.14	41.3
H-Bridges	Light goods vehicle	46	0.019	5.6
Motors	32 tonne (4 axle) truck	2.1e+02	0.037	11.0
Breadboard	32 tonne (4 axle) truck	2.1e+02	0.037	11.0
Arduino Mega	32 tonne (4 axle) truck	2.1e+02	0.037	11.0
Total		**2.1e+03**	**0.34**	**100**

Breakdown by components

Component	Mass (kg)	CO2 footprint (kg)	%
Kill Switch	0.05	0.0065	1.9
Battery	0.29	0.038	11.3
Servos	0.0069	0.00089	0.3
Motors	0.78	0.1	30.1
Ultrasonic Sensor	0.006	0.00078	0.2
Breadboard	0.06	0.0078	2.3
Arduino Mega	0.036	0.0047	1.4
Wires	0.005	0.00065	0.2
PLA	0.1	0.013	4.0
Wooden Dowels	0.75	0.097	28.8
H-Bridges	0.45	0.059	17.5
WifiCam	0.05	0.0065	1.9
Total	**2.6**	**0.34**	**100**

Use:

Static mode

Energy input and output type	Electric to mechanical (electric motors)
Country of use	North America
Power rating (W)	24
Usage (hours per day)	2
Usage (days per year)	15
Product life (years)	0.25

Relative contribution of static and mobile modes

Mode	**CO2 footprint (kg)**	**%**
Static	0.094	100.0
Mobile	0	
Total	**0.094**	**100**

Disposal:

Component	**End of life option**	**CO2 footprint (kg)**	**%**
Kill Switch	Reuse	0.0007	1.8
Battery	Reuse	0.0041	10.3
Servos	Reuse	9.7e-05	0.2
Motors	Reuse	0.011	27.4
Ultrasonic Sensor	Reuse	8.4e-05	0.2
Breadboard	Reuse	0.00084	2.1
Arduino Mega	Reuse	0.00051	1.3
Wires	Reuse	7e-05	0.2
PLA	Recycle	0.005	12.6
Wooden Dowels	Landfill	0.01	26.2
H-Bridges	Reuse	0.0064	15.9
WifiCam	Reuse	0.0007	1.8
Total		**0.04**	**100**

EoL potential:

Component	End of life option	CO2 footprint (kg)	%
Kill Switch	Reuse	0	0.0
Battery	Reuse	0	0.0
Servos	Reuse	-0.081	0.0
Motors	Reuse	-9.1	3.2
Ultrasonic Sensor	Reuse	0	0.0
Breadboard	Reuse	-0.31	0.1
Arduino Mega	Reuse	-0.35	0.1
Wires	Reuse	-0.034	0.0
PLA	Recycle	-0.15	0.1
Wooden Dowels	Landfill	0	0.0
H-Bridges	Reuse	-2.7e+02	96.4
WifiCam	Reuse	0	0.0
Total		**-2.8e+02**	**100**

COVER SHEET:

https://docs.google.com/document/d/1TBbpUobMKVg5YxTw_XNWbPypoomnfcW1xa-ZlMgxaA8/edit?usp=sharing